

## UAV Swing Payload Model

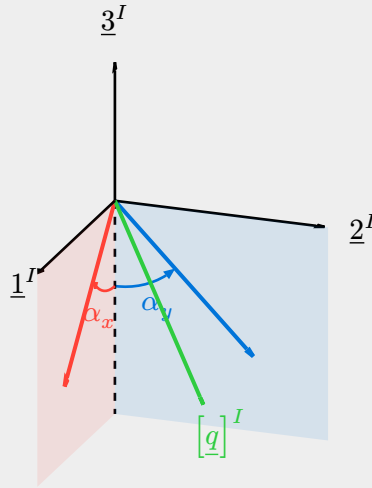
$$m_P \underline{a}_P = T \underline{q} - m_P \underline{g} \quad (1)$$

$$m_D \underline{a}_D = \underline{f} - T \underline{q} - m_D \underline{g} \quad (2)$$

$$\frac{d}{dt} \omega^{BE} = (J_B^B)^{-1} [\underline{n}_B - \underline{\Omega}^{BE} J_B^B \omega^{BE}] \quad (3)$$

$$\underline{s}_P = \underline{s}_D - L \underline{q} \quad (4)$$

### Swing angles



$$[\underline{q}]^I = \begin{pmatrix} \cos \alpha_x & 0 & -\sin \alpha_x \\ 0 & 1 & 0 \\ \sin \alpha_x & 0 & \cos \alpha_x \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha_y & -\sin \alpha_y \\ 0 & \sin \alpha_y & \cos \alpha_y \end{pmatrix} [-\underline{3}^I]^I = \begin{pmatrix} \sin \alpha_x \cos \alpha_y \\ \sin \alpha_y \\ -\cos \alpha_x \cos \alpha_y \end{pmatrix} \approx \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \quad (5)$$

## ArUco Marker Tracking EKF

### Frames

$[\cdot]^I$  : ENU (+x East, +y North, +z Up)

$[\cdot]^B$  : body (+x nose, +y portside, +z up) (6)

$[\cdot]^C$  : camera (+x right, +y up, +z optical axis)

### State

$$[\xi]^I = \begin{pmatrix} \alpha_x \\ \alpha_y \\ \dot{\alpha}_x \\ \dot{\alpha}_y \\ \psi_P \end{pmatrix} \in \mathbb{R}^5 \quad (7)$$

$$s_x = \sin \alpha_x, \quad c_x = \cos \alpha_x, \quad s_y = \sin \alpha_y, \quad c_y = \cos \alpha_y \quad (8)$$

$\alpha_x, \alpha_y$ : tether swing angles, tipping toward +x<sup>I</sup> (East) and +y<sup>I</sup> (North) [rad]

$\dot{\alpha}_x, \dot{\alpha}_y$ : swing rates [rad/s]

$\psi_P$ : payload yaw: marker-row direction [rad]

## Input

$$[T]^{IB} = \begin{pmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{pmatrix} \quad (9)$$

Euler angles:  $\varphi$ : roll  $\theta$ : pitch  $\psi$ : yaw

$$[\underline{a}_D]^I = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = [T]^{IB} [\underline{a}_D]^B \quad (10)$$

$[\underline{a}_D]^I$ : acceleration inputs from the drone [m/s<sup>2</sup>]

## Process model

$$\begin{bmatrix} \dot{\xi} \end{bmatrix}^I = f(\xi, a) = \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{c_x a_1 + s_x a_3}{L c_y} + 2 \frac{s_y}{c_y} \dot{\alpha}_x \dot{\alpha}_y \\ -\frac{c_y a_2 + s_y (c_x a_3 - s_x a_1)}{L} - s_y c_y \dot{\alpha}_x^2 \\ 0 \end{pmatrix} \approx \begin{pmatrix} \dot{\alpha}_x \\ \dot{\alpha}_y \\ -\frac{1}{L} (a_1 + \alpha_x \|g\|) \\ -\frac{1}{L} (a_2 + \alpha_y \|g\|) \\ 0 \end{pmatrix} \quad (11)$$

$L$ : drone tether-pivot to payload CG [m]

$$F = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\frac{\|g\|}{L} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\|g\|}{L} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (12)$$

$F \in \mathbb{R}^{5 \times 5}$ : process Jacobian (continuous-time)

## Prediction

Jacobian to discrete-time conversion w/ Euler integration.

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \Delta t f(\hat{\xi}_k^+, a_s) \quad (13)$$

$$\Phi_k = I_5 + F \Delta t, \quad Q = \text{diag}(0, 0, q_x, q_y, q_\psi) \Delta t \quad (14)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (15)$$

$\Delta t$ : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$ : estimate before & after the update at step  $k$

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$ : covariance before & after

$\Phi_k$ : discrete time Jacobian

$q$ : process-noise intensity

Shelving Jacobian to discrete-time conversion w/ RK4 for now.

$$k_1 = f(\hat{\xi}_k^+, a_s), \quad k_2 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_1, a_s\right) \quad (16)$$

$$k_3 = f\left(\hat{\xi}_k^+ + \frac{\Delta t}{2} k_2, a_s\right), \quad k_4 = f(\hat{\xi}_k^+ + \Delta t k_3, a_s)$$

$$\hat{\xi}_{k+1}^- = \hat{\xi}_k^+ + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (17)$$

$$\Phi_k = I_5 + F \Delta t + \frac{1}{2} (F \Delta t)^2, \quad Q = \text{diag}(0, 0, q, q, q_\psi) \Delta t \quad (18)$$

$$P_{k+1}^- = \Phi_k P_k^+ \Phi_k^\top + Q \quad (19)$$

$\Delta t$ : time step [s]

$\hat{\xi}_k^-, \hat{\xi}_k^+$ : estimate before & after the update at step  $k$

$P_k^-, P_k^+ \in \mathbb{R}^{5 \times 5}$ : covariance before & after

$\Phi_k$ : discrete time Jacobian

$q$ : process-noise intensity

## Measurement

### Pre-processing logic

For every  $j \in V_k$ , solvePnP returns the marker pose in the camera frame,

$$[t_j]^C, [T]^{CM_j} \quad j \in V_k \quad (20)$$

$V_k$ : marker IDs detected in frame  $k$

$[t_j]^C$ : marker  $j$  center in the camera frame [m]

$[T]^{CM_j}$ : transformation to the camera frame from the marker local frame  $M_j$

The first column of  $[T]^{CM_j}$  is the marker  $x$ -axis, which runs along the marker row,

$$[m_j]^C = [T]^{CM_j} [\underline{1}^{M_j}]^{M_j} \quad (21)$$

Each marker gives its own estimate of the board center; average over the markers actually detected in the frame,

$$[p_{\text{ctr}}]^C = \frac{1}{m_k} \sum_{j \in V_k} \left( [t_j]^C + o_j [m_j]^C \right) \quad (22)$$

$[p_{\text{ctr}}]^C$ : payload center in the camera frame

$m_k$ : # of markers in the frame

$o_j$ : marker offset from board center along the row, +left [m]

Shift the origin from the camera optical center to the tether pivot,

$$[p]^C = [p_{\text{ctr}}]^C + [T]^{CB} ([t_{BC}]^B - [\ell]^B) \quad (23)$$

$[p]^C$ : payload center after the camera offset is corrected, camera frame [m]

$[t_{BC}]^B$ : camera optical center in the body frame [m]

$[\ell]^B$ : drone tether pivot in the body frame [m]

Average the payload yaw and transform into the inertial frame,

$$[m]^I = [T]^{IB} [T]^{BC} \left( \frac{1}{m_k} \sum_{j \in V_k} [m_j]^C \right) \quad (24)$$

$$z_{\psi_P} = \text{atan2}(m_2^I, m_1^I)$$

### Measurement model

After some pre-processing from pixels to camera frame coordinates, we measure,

$$z_k = \begin{pmatrix} \frac{p_x^C}{\|p\|} \\ \frac{p_y^C}{\|p\|} \\ z_{\psi_P} \end{pmatrix} + \varepsilon_k \quad (25)$$

$p$ : payload center position

$z_{\psi_P}$ : measured payload yaw

$\varepsilon_k$ : measurement noise

### Prediction of measurement

Measurement predicted from swing angles,

$$h(\xi) = \begin{pmatrix} q_x^C \\ q_y^C \\ \psi_P \end{pmatrix} \quad [\underline{q}]^C = [T]^{CB} [T]^{BI} \begin{pmatrix} \alpha_x \\ \alpha_y \\ -1 \end{pmatrix} \quad (26)$$

$$H = \frac{\partial h}{\partial \xi} = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} [T]^{CB} [T]^{BI} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ (0 \ 0 \ 0 \ 0 \ 1) \end{pmatrix} \quad (27)$$

## Update

### Innovation

$$y_k = z_k - h_k(\hat{\xi}_k^-) \quad (28)$$

$$R_k = \text{diag}(\sigma_x^2, \sigma_y^2, \sigma_\psi^2) \in \mathbb{R}^{3 \times 3} \quad (29)$$

$$S_k = H_k P_k^- H_k^\top + R_k \in \mathbb{R}^{3 \times 3} \quad (30)$$

$y_k$ : innovation

$R_k$ : measurement noise

$S_k$ : innovation covariance

### Kalman gain

$$K_k = P_k^- H_k^\top S_k^{-1} \in \mathbb{R}^{5 \times 3} \quad (31)$$

$K_k \in \mathbb{R}^{5 \times 3}$ : Kalman gain

$$\hat{\xi}_k^+ = \hat{\xi}_k^- + K_k y_k, \quad (32)$$

$$P_k^+ = (I_5 - K_k H_k) P_k^- \quad (33)$$

If no markers detected:

$$\hat{\xi}_k^+ = \hat{\xi}_k^-, \quad P_k^+ = P_k^- \quad (34)$$